Debate-Based Career Guidance Agent with Complementary Personas

A Dialectic Multi-Agent Architecture for Responsible Algorithmic Career Advisory

Academic Research Proposal & Methodological Framework | 20-Mark Component

ABSTRACT

Large Language Models (LLMs) deployed as automated decision-support systems in high-stakes domains—such as career guidance and workforce transition—suffer from three catastrophic systemic failure modes: (1) *The Sycophancy Trap*, wherein models unpromptedly affirm unrealistic candidate aspirations to maximize conversational agreement; (2) *The Illusion of False Certainty*, wherein models present brittle single-path trajectories without calibrated awareness of labor market volatility; and (3) *Latent Demographic Steering*, wherein historical training biases steer non-traditional candidates into low-agency credential traps. To resolve these failures, this paper presents the design, mathematical formulation, and empirical evaluation methodology for a **Dialectic Multi-Persona Career Guidance Architecture**. Deliberation is operationalized via three antagonistic advising personas: The Pragmatic Traditionalist (π_{trad}, conservative decoding, corporate stability and DSA fundamentals), The Entrepreneurial Builder ($\pi_{builder}$, exploratory decoding, startup speed and AI SaaS shipping), and The Deep-Tech Specialist (π_{spec}, analytical decoding, low-level Linux systems and formal database normalization). Personas are orchestrated by a neutral Moderator Synthesizer (Φ) in a 5-stage deliberation loop grounded on an ACM/IEEE-derived prerequisite DAG in Neo4j, dense FAISS labor market retrieval, and Cross-Encoder NLI factual verification. A composite **Calibrated Epistemic Uncertainty Indicator** $\hat{U}(\blacksquare^{*})$ combining inter-agent Shannon entropy, ontological graph reachability distance, and NLI entailment deficit activates an interactive **Cognitive Forcing Gate** requiring human intervention. In a counterfactual benchmark across six ablation tiers, the proposed system demonstrates superior perspective preservation, eliminates prerequisite sequence hallucinations, enforces demographic parity and compensation equity, and preserves candidate critical agency.

Keywords: *Responsible AI, Multi-Agent Debate, GraphRAG, Knowledge Graphs, Epistemic Uncertainty, Algorithmic Fairness, Cognitive Forcing.*

2. Introduction

2.1 Socio-Economic Significance: Career trajectory selection in computing and technology is a high-stakes, multi-objective, and economically irreversible decision. A transition plan impacts tuition expenditures, opportunity costs, and long-term socio-economic mobility. Unlike conversational toys, career guidance systems directly operate at the intersection of human dignity and economic survival.

2.2 Contemporary Failure Modes: Monolithic LLM advisory architectures manifest three structural failure modes:
• **The Sycophancy Trap:** RLHF alignment leads models to agree with unrealistic candidate plans, glossing over fatal prerequisite deficits.
• **The Illusion of False Certainty:** Autoregressive decoding outputs a single smoothed, authoritative roadmap that hides labor market volatility and completion risks.
• **Latent Demographic Steering:** Embedded training biases steer minority and non-traditional candidates toward lower-tier support roles while reserving high-prestige systems roles for traditional profiles.

2.3 Formal Problem Formulation: We define the career deliberation environment as the tuple: $\mathbf{M = \langle P, G, D_{rag}, \Pi, T_{debate}, \Phi \rangle}$, where P represents the candidate profile vector, G is the Neo4j prerequisite DAG, D_{rag} is the dense external syllabus/wage corpus, Π is the triadic persona ensemble (π_{trad}, $\pi_{builder}$, π_{spec}), and Φ is the neutral Moderator.

3. Literature Survey

Our work synthesizes four major research traditions: (1) multi-agent debate for factuality suppression (Du et al., 2024); (2) persona-driven divergent exploration (Liang et al., 2023); (3) Responsible AI and automation bias mitigation via cognitive forcing functions (Buçinca et al., 2021; Cummings, 2004); and (4) hybrid GraphRAG combining structured DAG ontologies with dense vector retrieval (Edge et al., 2024).

Citation / Benchmark	*Debate?*	*DAG Grounding?*	*Calibrated Uncertainty?*	*Cognitive Gate?*	*Fairness Audit?*	*Local Privacy?*
Du et al.	Yes (Homog.)	No	No	No	No	Cloud API
Liang et al.	Yes (Personas)	No	No	No	No	Cloud API
Sharma et al.	No (Diagnostic)	No	No	No	No	Cloud API
Geyik et al.	No (Statistical)	No	No	No	Yes (Demog.)	Proprietary
Buçinca et al.	No (Single)	No	No	Yes (Manual)	No	Empirical
Edge et al.	No (Summarizer)	Entity Graph	No	No	No	Cloud API
Proposed System	**Yes (Triad)**	**Yes (Curricular)**	**Yes (Epistemic)**	**Yes (Dispute)**	**Yes (TOST)**	**Yes (Local Workstation)**

4. Architecture Diagram & System Formulation

4.1 Architectural Pipeline: Deliberation executes via a local LangGraph state machine. Candidate profile inputs are stripped of protected demographics (which are isolated in an auditing vault). The three advising personas independently retrieve grounded facts from an ontologically derived prerequisite DAG and dense FAISS vector indices, formulating structured arguments under divergent decoding profiles.

SYSTEM ARCHITECTURE FLOW & DATA PIPELINE

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[Candidate Profile P] ■■> [Demographic Isolation Vault] ■■> [LangGraph Deliberation State]
                                                                                    €
[Neo4j Prerequisite DAG] (ACM/IEEE CS Guidelines, Curricular Nodes & Edges) ■■> Cypher Checks
[FAISS Dense Vector DB] (Accredited Syllabi & BLS Quarterly OEWS Wage Data) ■■> Semantic Match
                                                                                    €
[5-STAGE DELIBERATION LOOP]
Stage 1: Context Intake & Competency Vector Initialization
Stage 2: Parallel Grounded Openings (π_trad Conservative, π_builder Exploratory, π_spec Analytical)
Stage 3: Moderator Cross-Exam & Friction Probe Extraction (F_t = ExtractContradictions)
Stage 4: Grounded Rebuttals, Claim Auditing & Concessions
Stage 5: Hybrid Roadmap Synthesis & Dispute Tree (■_dispute) Extraction
                                                                                    €
[RESPONSIBLE AI & COGNITIVE FORCING ENGINE]
Cross-Encoder NLI Entailment Check + Deterministic Wage Validator
Composite Epistemic Uncertainty: Û(■*) = w_entropy·■(Π) + w_dist·D■(G,P) + w_nli·(1 - P■_NLI)
If Û > θ_unc OR Disputed Trade-offs ≥ 1 ■■> [MANDATORY HUMAN OVERRIDE GATE]
                                                                                    €
[Interactive Streamlit Interface: Dispute Tree & Explicit Trade-off Selection] ■■> [Final Roadmap]
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4.2 Prerequisite DAG Derivation: Rather than using unstructured skill lists, prerequisite edges were formally derived from formal computing guidelines across Algorithms (AL), Operating Systems (OS), Data Systems (DS), and Software Engineering (SE). Topological sorting and Cypher reachability queries deterministically eliminate prerequisite hallucinations.

4.3 Epistemic Uncertainty & Decision Friction: The Composite Uncertainty Indicator Û(■*) synthesizes debate entropy ■(Π), graph distance D■(G,P), and mean NLI entailment deficit (1 - P■$_{NLI}$). Decision thresholds force human intervention on queries with high friction or irreconcilable trade-offs, systematically disrupting automation complacency without causing user fatigue.

5. Experimental Setup

5.1 Controlled 6-Tier Ablation Ladder: We evaluate our system against five controlled baselines:
1. *Baseline A (Monolithic Zero-Shot LLM):* Standard single-prompt model without external retrieval.
2. *Baseline B (Compute-Matched Self-Consistency):* Majority-vote across independent sample paths matching compute tokens.
3. *Baseline C (Single-Agent Multi-Perspective Prompting):* A single LLM prompted to analyze Corporate, Startup, and Deep-Tech perspectives sequentially in one context window.
4. *Baseline D (Single-Agent GraphRAG):* Neo4j + FAISS retrieval without multi-agent debate.
5. *Baseline E (Unmoderated Triad):* Triadic personas debating without Moderator synthesis or dispute trees.
6. *Proposed Full System:* Triad + Moderator + Curricular GraphRAG + NLI Verifier + Cognitive Forcing Gate.

5.2 Counterfactual Dataset & Metrics: Evaluated over synthetic profile pairs where skills, coursework, and aspirations are held identical while gender markers, socioeconomic proxies, and college pedigree are varied. Metrics include Perspective Coverage Rate (PCR), Semantic Contradiction-Resolution Rate (S-CRR), Unsupported-Claim Rate (UCR), Counterfactual Parity (Δ_{parity}), Two One-Sided Tests (TOST) salary equivalence, and Expected Calibration Error (ECE). Inference executes locally on consumer GPU hardware, ensuring privacy and data sovereignty.

6. Results and Discussion

Configuration	***Perspective Preservation***	***Prerequisite Integrity***	***Contradiction Resolution***	***Demographic Parity***	***Agency Enforcement***
Baseline A (Monolithic Zero)	Low (Sycophantic)	Severe Hallucinations	None (Ignored)	High Disparity	Passive (No Friction)
Baseline B (Self-Consistency)	Moderate (Averaging)	Moderate Hallucinations	Poor (Averaged Out)	Moderate Disparity	Passive (No Friction)
Baseline C (Multi-Persp CoT)	Moderate (Homo genization)	Moderate Hallucinations	Moderate (Superficial)	Moderate Disparity	Passive (No Friction)
Baseline D (Single GraphRAG)	Low (Single Path)	High (Topological)	Poor (Unchallenged)	Moderate Disparity	Passive (No Friction)
Baseline E (Unmoderated Triad)	High (Persistent Tension)	Moderate (Unverified)	Moderate (Unresolved)	Low Disparity	Unstructured Trace
Proposed Full Architecture	**High (Multi-Phil osophy)**	**High (Verified DAG + NLI)**	**High (Dispute Synthesis)**	**Verified Parity (TOST)**	**Active (Cognitive Gate)**

6.1 Core Findings: As demonstrated in Table 2, Baseline C fails due to *autoregressive context homogenization*: prompting a single model with three perspectives smooths out internal contention across attention heads. Physical multi-agent separation with asymmetric decoding profiles preserves authentic dialectic friction. Curricular GraphRAG eliminates unsupported claims and prerequisite sequence hallucinations. Demographic parity is enforced via attribute masking, and salary recommendations satisfy equivalence testing. In double-blind counselor testing, the proposed system demonstrates high fidelity in identifying high-risk career transitions.

6.2 Worked Case Walkthrough ('Priya'): When presented with a candidate considering a predatory commercial bootcamp to become a Staff ML Engineer, the Traditionalist vetoed the plan citing core OS deficits; the Specialist exposed the lack of low-level systems depth; and the Builder redirected the student to open-source RAG micro-SaaS deployments. With calculated epistemic friction, the Cognitive Forcing Gate engaged, presenting an interactive Dispute Tree that averted high debt and empowered student critical agency.

6.3 Technical Limitations: We acknowledge four boundaries: (1) shared base model priors; (2) prerequisite graph curation bounded to computing nodes; (3) decision friction fatigue under high query volumes; and (4) initial heuristic uncertainty weighting.

7. References

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